# Global Talent Migration

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***Abstract*—This project aims to analyse and identify trends of migration in relation to country, industry and skill. The LinkedIn - World Bank partnership has created a comprehensive dataset that tracks the movement of LinkedIn members worldwide. Three datasets have been used for analysing talent migration across countries, industries and skills. Docker is used to run a choice of databases which is Mongo DB and PostgreSQL to store and manage data. Data analysis and visualisation are done on Jupyter Notebook using Python. Preprocessing, transformation, analysis and visualisation tasks are performed using Python libraries such as pandas, numpy, matplotlib and seaborn. To build and manage a robust and reliable data pipeline Dagster is used. ETL (Extract, Transform, Load) process for datasets followed by storing results in PostgreSQL is performed using Dagster.**

***Index Terms*—python, dagster, skill, country, industry**

## I. Introduction

A skilled workforce is the foremost criterion of migration across different countries and industries, therefore, it is imperative to find the particular skill and industry in which a country can grow economically. Country migration takes place due to better work opportunities and standard of living. The motivation of this project is to examine the worldwide patterns of migration in relation to country, industry, and skill, in order to produce insights for decision-makers and analysts regarding which area has a more significant influence on the development of an industry or country. Additionally, the findings from this study provide insights into skill penetration, employment shifts and talent migration in over 100 countries across various LinkedIn members.

With its vast professional network, LinkedIn is an indispensable tool for talent migration, connecting job seekers with employers globally. As the world continues to globalize, its role in facilitating professional connections will become increasingly important. The insights provided by country, skill and industry migration data will enable leaders to identify areas of opportunity and make informed decisions to benefit their workforce and economy.

The metrics of talent migration in the datasets are based on net migrations, that is, arrival minus departure from countries, this gives fair comparisons over different countries, industries and skills across various LinkedIn members. Our choice of the database has been Mongo DB, as it allows scalability and flexibility, thus, making it easy to add new data to the database as needed. Jupyter Notebook is used for data analysis and visualisation using Python. Dagster a data orchestration platform and Postgre SQL is used to build and manage a robust and reliable data pipeline.

## II. Related Work

Skilled migration benefits innovation, especially in technology-focused industries with an openness to international trade. According to the authors skilled migrant workers have a positive impact on innovation in European countries by bringing new ideas and knowledge [1]. For a better understanding of skill in the context of cross-border labour mobility, the authors highlight the need for a more nuanced comprehension of its social construction. They advise policymakers to consider this aspect when creating labour migration and skill recognition policies. Furthermore, the authors recommend that future research concentrate on the intersection of skill, power dynamics, and cultural values concerning cross-border labour mobility [2].

Workers migrate to improve job opportunities and living standards, but mobility can be hindered by market and regulatory failures. Benefits for both regions depend on job outcomes. Migration can promote local development through remittances and investment, but without local investments, it can worsen income inequality and social instability in origin regions. [3]. Industrial relocation has a major impact on regional and local wealth generation. Policymakers must understand the factors that govern it. In the Netherlands and the UK, timing of migration is linked to business cycle fluctuations, and sector choice is linked to transportation reliance and operation size. Municipalities are chosen based on industrial real estate availability, cost, local amenities, and proximity to the city center. [4].

Migration and development are linked globally, with remittances supporting wealth in origin nations. Migration patterns vary due to global economic changes, internal economic reforms, and socio-political transformations. Skilled migration has benefits and drawbacks in the short and long run for both origin and destination countries. The focus is on how the economies of significant sending regions are affected by outflow of highly skilled workers [9].

Existing literature reveals a gap in analyzing and visualizing global talent migration, including skills, country, and industry of origin. Country policies impact talent migration and industry growth by attracting a talent pool. Our study analyzes global talent migration from 2015-2019, focusing on skills, industries, and countries. The analysis aims to identify top-performing skills, countries, and industries that impact global decision-making.

## III. Methodology

In today's world, industries rely on databases and analytics, including global talent migration. With LinkedIn and World Bank datasets, organizations can use Jupyter Notebook, Python, Dagster, Mongo DB, and PostgreSQL to process, analyze, and visualize data. An open-source web application, Jupyter Notebook is used to combine code, visualizations, and text to provide an interactive way to explore and present data. A programming language such as Python, having a rich set of libraries and frameworks for data manipulation and analysis is used for data cleaning, data transformation, and data visualization. Python's powerful libraries such as NumPy, Pandas, and Matplotlib provided the necessary tools to explore and manipulate large datasets effectively.
Dagster is selected for building ETL (extract, transform, load) pipelines. It provided a unified view of the entire data pipeline and simplified the coordination of complex workflows. Dagster ensured data integrity and reliability which allowed us to schedule, run, and monitor complex data processing pipelines, therefore, the final output is reliable and accurate. PostgreSQL, an object-relational database management system is used due to its reliability, scalability, and provision of a secure platform for storing and managing data. Its robustness and reliability made it an excellent choice for global talent migration data project. MongoDB is selected as it is a type of NoSQL database that utilizes document-based data storage to accommodate vast amounts of data, along with robust indexing features and efficient query engines [5].

The LinkedIn - World Bank dataset is chosen to perform analysis for country, industry and skill migration globally. The migration of talent between countries and within them is determined by the location information provided by users. If a user changes their job location to a different location, this is considered a physical move by LinkedIn.

The net gain or loss of members from a foreign country is calculated as a ratio of the average LinkedIn membership of the selected country during a given time period, divided by the result of this division is then multiplied by 10,000. To calculate the net migration between countries at a given time having a base country from which migration takes place to the target country is done using below formula:

$$CountryNetMigration = \frac{NetFlows}{MemberCount} * 10,000 \quad (1)$$

The industries that experienced growth or decline are determined by the industry of a member's company at the time of their migration. The net gain or loss of members from a foreign country in a particular industry is calculated as a ratio of the number of LinkedIn members working in that industry in the selected country during a specific period, divided by this result is then multiplied by 10,000. To calculate the net migration between industries in different countries at a given time is done using below formula:

$$NetIndustryMigration = \frac{NetIndustryFlows}{MemberIndustryCount} \quad (2)$$

The acquisition or loss of skills is determined by the skills listed on a member's profile at the time of their migration. The net gain or loss of members from a foreign country with a particular skill is calculated as a ratio of the number of LinkedIn members in the selected country who possess that skill during a specific period, divided by this result is then multiplied by 10,000. To calculate the net migration of skills in different countries at a given time is done using below formula:

$$NetSkillMigration = \frac{NetSkillFlows}{MemberSkillCount} \quad (3)$$

Three datasets of country migration, industry migration and skill migration are selected to perform data processing. The following sub-section provides processes performed on the three datasets.

### A. Country Migration

Country migration dataset contains 18 columns with no missing data. Each row represents a pair of countries such as the base country and target country which shows migration from one to another. Additional columns are country code, country latitude and longitude, World Bank income category, World Bank region, and internet penetration rate per 10,000 people for each year from 2015 to 2019.

To analyse the dataset and draw results, various visualization techniques and manipulations are used. A tableName variable is defined for country_migration data. A connection to the database is established and with the use of pandas, data is read from the specified table into a pandas dataframe called country_migration_data. This prints the sum of missing values in each column of the dataframe and creates a new dataframe by grouping and sorting the data by country name and net migration rates. The top 10 countries with the highest net migration rates are plotted in a bar chart using Plotly. A scatter plot is created which compares net migration rates from 2015 to 2019. Two dataframes are created and grouped, one with top 10 target countries by count and another with top 10 values based on the count.

A table is created with the top 10 target countries and the number of immigrants, and a choropleth map is created to display the number of immigrants per 10,000 people for each country in each year from 2015 to 2019. The map is also animated and displays the changes in migration trends over time. The map is based on GeoJSON data for the world map.

The analysis gives the result that the United States of America is the most preferred country for migration. UAE and Luxembourg have seen the highest amount of people leaving the country in 2015 i.e. 1713 and 1493 per 10k people. This count is reduced to 576 for UAE and 1107 for Luxembourg. Net Migration by Base country income shows that there is an increase in migration of people having lower middle Income and High Income. This may be due to better pay and job security.

### B. *Industry Migration*

Industry migration data contains columns of country code, country name, world bank defined income, world bank defined region, isic section index, isic section name, industry id, industry name and net migration per 10,000 for five years - 2015, 2016, 2017, 2018, 2019. Jupyter Notebook created for this dataset contains imported Python libraries such as pandas to load a dataset into a data frame, numpy to perform numerical computations, matplotlib to create a plot of the data and seaborn to enhance the plot with additional information, such as colour coding, statistical annotations, or facet grids. CSV file named industry_migration_public is loaded into a pandas DataFrame called industry_migration_data. Data is explored which gives us information such as the head of the data, number of rows, number of columns, missing values and type of different values present in our dataset.

Data transformation includes dropping null values, identifying unique values, checking for duplicates, and dropping unnecessary code columns. Grouping and sorting are performed for visualizing net migration based on country, industry, and isic section. The melt and reset index method is used to make the data plot easier. Bar plots display top industries, countries, and isic sections with the highest net migration each year. The top 10 industries are selected using the head method based on sorted values.

Data visualisation of country, income, region, industry and isic section with net migration gives us the result that Luxembourg is a high-income country in Europe and Central Asia, is growing from 2015 to 2019 in the Internet industry with information and communication isic section. While Qatar a high-income country in the Middle East and North Africa region is experiencing a massive decline in the transportation/trucking/railroad industry with transportation and storage isic section.

### C. *Skill Migration*

Skill migration dataset contains columns of country code, country name, world bank defined income, world bank defined region, skill group id, skill group category, skill group name and net migration per 10,000 for five years - 2015, 2016, 2017, 2018, 2019.
The skill group ID, category and name are provided to identify the type of skills being migrated. This information can be used to analyze the patterns and trends of skill migration across different countries and regions over time. The data is loaded in the pandas dataframe to perform cleaning using for loop and column iterations to remove null columns. To draw insights from cleaned data, visualisation is performed. Data visualisation shows which skills are lost or gained based on country migration and which skill group has better migrations.

The results of the analysis reveal a discernible trend, wherein migrants from low-income and lower-middle-income countries tend to migrate to high-income countries, thereby leading to a dearth of skilled labour in their respective countries. Furthermore, the findings indicate that countries classified as High Income continue to attract an increasing number of migrants annually. A significant skill migration from Afghanistan is observed and on the other hand, UAE has the least. From the regional point of view, North America, Europe and Central Asia have gained the most number of migrants, and Latin America and the Caribbean have the most number of migrants. East Asia and the Pacific region have also gained a large number of migrants.

## IV. Results and Evaluation

As the world becomes more interconnected, the rewards for linking skilled individuals with fitting employment or prospects are becoming increasingly substantial. These international talent exchanges hold extensive implications that affect a range of areas, spanning from industries to the country's economy [6].
For the choice of a pipeline, Dagster was used which performed a set of operations that are executed in a sequence to transform data from one form to another. Each operation in a pipeline takes input data and returns output data. Here dagster pipeline is used to build complex data pipelines by breaking them down into smaller components. It enabled a provision for a flexible and easy-to-use platform for building, testing, and managing data pipelines.

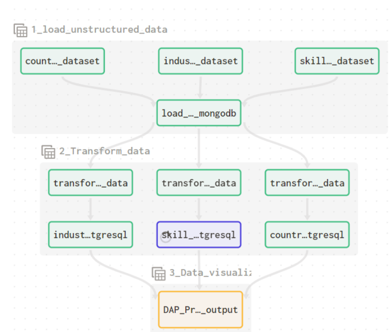


Fig. 1. Data Pipeline using Dagster

The results of country migration data showed that the United States is the top destination for highly skilled talent from other countries, with over 3 million migrant workers in the LinkedIn dataset. Other top destinations include the United Kingdom, Canada, and Australia. The top source countries of highly skilled talent are India, the United Kingdom, and Canada. These countries account for a large share of the highly skilled migrants in the dataset.

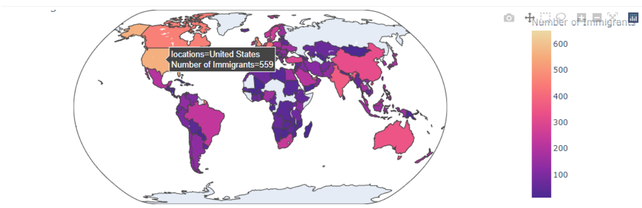


Fig. 2. Country Growth by Net Migration

The results of industry migration data showed that the most popular destination for the Internet industry is Luxembourg, followed by Qatar for the Food Production industry. Furthermore, there is a significant drop in the transportation / trucking / railroad industry in Qatar from 2015 to 2019. From 2015 to 2019 the graph shows the comparison between different countries, industries, isic section and region defined by the world bank. While highly skilled migration can bring benefits to both sending and receiving industries and countries, it can also lead to imbalance and brain drain in the sending countries, as they lose their most highly educated and skilled workers for the particular industry which holds the potential to grow the country's economy.

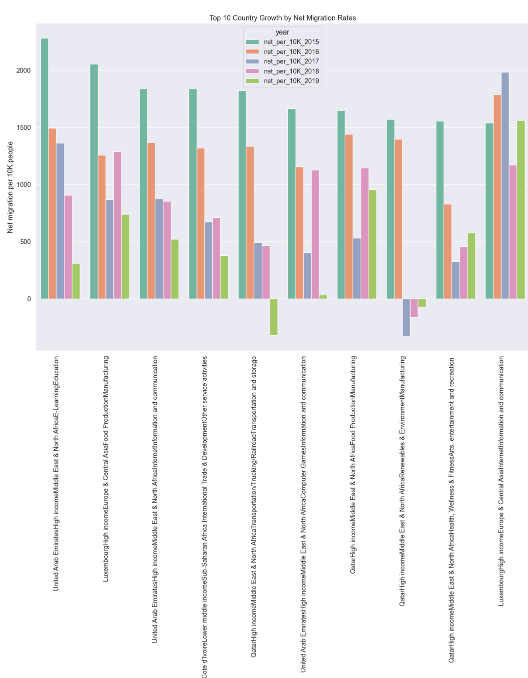


Fig. 3. Top 10 Country and Industry Growth by Net Migration

The technology industry is the leading industry for both sending and receiving highly skilled talent, with over 40% of all highly skilled talent in the dataset working in technology-related fields. Interestingly, the data also showed that there is a huge increase in specialised industry skills in 2019 followed by a business skills increase. There is a steady increase in all skills from 2015 to 2019, however, the software development life cycle (SDLC) skill group has the most increase compared to data storage technologies which stands at second in growth.

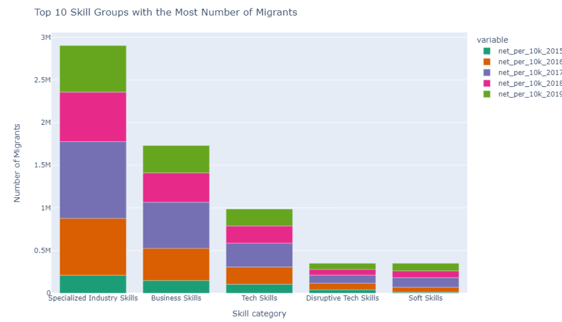


Fig. 4. Top Skill Category with Net Migration

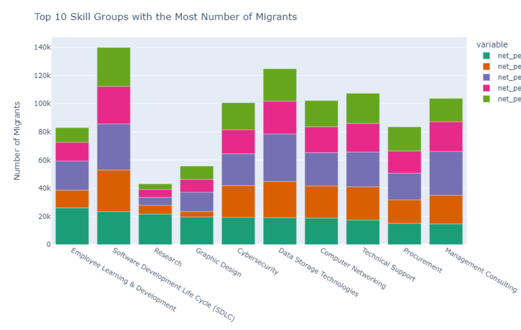


Fig. 5. Top Skilled Group with Net Migration

## V. Conclusion and Future Work

World Bank and LinkedIn data show global talent migration affects countries losing and gaining talent in different industries. Losing talent risks losing competitiveness, while gaining talent has potential for growth and innovation. Policymakers must develop strategies to retain skilled workers and attract international talent. It is withdrawn that Jupyter Notebook, Python, Dagster, and PostgreSQL technologies provided a unified platform for data processing, analysis, and visualization, which enabled us to explore complex datasets effectively.

Based on our project analysis, India is the top-performing country in terms of migration into the country, potentially due to the pandemic and the return of the workforce to their home country. Conversely, UAE experienced migration out of the country, while Afghanistan saw more people leaving for better job opportunities. The E-learning industry experienced rapid growth in 2019, potentially due to the global shift towards online work and learning, while education saw a decline in migration, possibly due to a decrease in offline teaching opportunities. It is of paramount importance to acknowledge that our dataset is predicated on the analysis of information updates made by LinkedIn members. It should be noted that there may be more migrations from one country or industry to another, influenced by various factors. However, the dataset in question only reflects the online presence and a limited number of population migrations, as compared to the actual figures.

The COVID-19 pandemic, economic recessions, and the rise of populism have led to tensions between policies that prioritize citizen skillsets and livelihoods and those that attract key sector essential workers with fluid definitions [7]. It is noteworthy that the datasets employed for our analysis were sourced exclusively from LinkedIn member information updates spanning the period of 2015 to 2019. While our findings align with this time frame, it is essential to acknowledge that the global pandemic of 2020 brought about significant changes. Post pandemic world witnessed digital transformation procedures that influenced three aspects of business operations, specifically labour and social relations, marketing and sales, and technology. Therefore, to gain a comprehensive understanding of the present migration patterns, it would be advantageous to incorporate data from the post-2020 period. The adoption of new work models also drives the need for fresh talent, regardless of their location. Additionally, with digitalisation cybersecurity and privacy are seen as critical factors that contribute to the unified growth of the Internet of Things technology solutions, artificial intelligence, big data, and robotics [8].

Furthermore, it is crucial to acknowledge that the datasets reflect only a fraction of the population migration and are based on the online presence of LinkedIn users. Despite these limitations, data analytics presents a powerful tool for countries, individuals, and industries to make informed decisions and formulate policies that promote economic growth and development.

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